

Introduction to Microeconomics

Lecture 15—Costs

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ECON101
Fall 2023

Costs of Production: Which costs matter?

Economic Costs vs Accounting costs

Accounting Costs Cost of an activity, as recorded. Tangible costs comprising fundamentals of running a firm: payroll, production costs, marketing, capital depreciation charges, etc. In general: Accounting costs = manufacturing costs + labor, salaries, taxes + additional expenses.

Economic Costs explicit costs (accounting) plus **implicit costs OR cost of opportunity**

- the cost of resources that the firm used that could have been put to some other use. For example, an entrepreneur who owns a business could use her time to earn income at a salaried job.

Total Costs, Variable Costs, Fixed Costs and Sunk Costs

Cost Taxonomy

Total Cost Total economic cost of production and it is the sum of **Sunk Costs (fixed)**, **avoidable fixed costs** and **Variable Costs (avoidable)**

Variable Costs These are costs that vary with the level of output. These costs can be avoided by not producing.

Fixed Costs Can be further subdivided in two

Sunk Costs Costs that are irrevocably committed over the relevant time period, thus **cannot be avoided**

Avoidable fixed costs Costs that do not vary with the level of output, but that can be avoided by not producing.

Sunk and avoidable costs are concepts related to time and timing, which is a dynamic issue. A cost is fixed if it is independent of the level of output, whereas it is variable if it varies with the production scale. Fixed and variable costs are then concepts determined by technology and the legal system. These distinctions are important for the planning timing of the firm under consideration, as well as the shutdown and production decisions of the firm!

More about costs

- 1 Total costs can be categorized as

$$\begin{aligned}\text{Total cost} &= \text{Sunk costs} + \text{Avoidable costs} \\ &= \text{Sunk costs (Fixed)} + \text{avoidable fixed costs} + \text{Variable costs (avoidable)} \\ &= \text{Fixed Costs} + \text{Variable Costs}\end{aligned}$$

- 2 Fixed costs consist of sunk costs and avoidable fixed costs.
- 3 Avoidable costs include avoidable fixed costs and variable costs.
- 4 In the long run, **there are no sunk costs**, but there may easily be fixed costs.
- 5 In the short run, at least some costs are sunk, and there may still be avoidable fixed costs.

- Production function: $q = F(K, L)$, how much q given K, L ?
- Cost Function: $C(q, \text{input prices})$, how costly is to produce q given some inputs?
 - With two inputs, K and L , $\text{Costs} = wL + rK$, where w is the wage per hour and r is the cost of capital.
- The goal of the firm is to choose inputs optimally, in the cheapest possible way, so as to produce q output.

Long Run costs

- There are no sunk costs, so in the planning problem of the firm, all inputs can be altered freely and chosen optimally to minimize cost given a target level of output.

Long Run Cost minimization problem

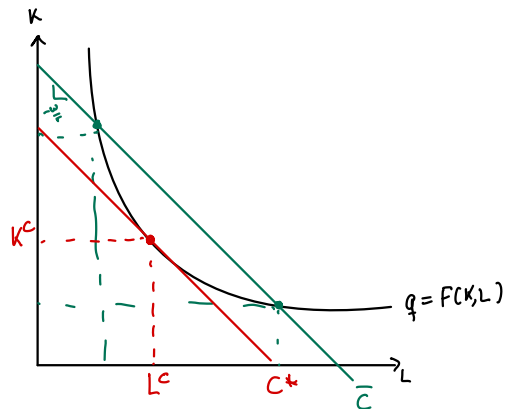
$$\min_{K,L} wL + rK \quad \text{subject to: } q = F(K, L)$$

In this problem, input prices w and r , as well as q are treated as parameters or **exogenous variables**. The Technology F is also fixed. The optimal solutions for K and L are **endogenous variables**.

- We call the optimal solutions the **conditional input demands**: $K^c(w, r, q)$, $L^c(w, r, q)$.
- There is a dual relationship between the production function and the cost function. In problems, the cost function is sometimes used as a primitive, instead of the production function.
- This problem is actually mathematically the same as the consumer's expenditure minimization problem.

Graphical Analysis and the Optimality Condition

$$\min_{K, L} \quad wL + rK : \text{ s.t. } q = F(K, L)$$



Iso-cost line: $\bar{C} = wL + rK$
where $\bar{C}$ is a fixed level of total costs

$$K = \frac{\bar{C}}{r} - \left(\frac{w}{r}\right)L$$

$\frac{w}{r}$ = input price ratio

Optimality Condition (Interior Optimum)

$$MRTS = \frac{MP_L}{MP_K} = \frac{w}{r} \Leftrightarrow \frac{MP_L}{w} = \frac{MP_K}{r}$$

$$\frac{MP_L}{MP_K} = \frac{w}{r}, \text{ solve with } q = F(K, L)$$

$$\Rightarrow K^c(w, r, q), L^c(w, r, q)$$

$$C^* = C(w, r, q) = wL^c(w, r, q) + rK^c(w, r, q)$$

Long-Run Total Cost Function.

Example: $q = A K^\alpha L^\beta$ ($\sigma = 1$)

$$\begin{aligned} MP_L &= A^\beta K^\alpha L^{\beta-1} \\ MP_K &= A^\alpha K^{\alpha-1} L^\beta \end{aligned} \Rightarrow MRTS = \frac{MP_L}{MP_K} = \frac{\beta}{\alpha} \frac{K}{L}$$

Long-Run Cost Minimization

$$\begin{aligned} \min_{K, L} \quad & \omega L + rK \\ \text{s.t.} \quad & q = A K^\alpha L^\beta \end{aligned}$$

Opt. Cond. $\frac{\beta}{\alpha} \frac{K}{L} = \frac{\omega}{r} \Rightarrow K = \frac{\alpha}{\beta} \frac{\omega}{r} L$, plug into $q = A K^\alpha L^\beta$

$$\Rightarrow q = A \left(\frac{\alpha}{\beta} \frac{\omega}{r} \right)^\alpha L^\alpha L^\beta = A \left(\frac{\alpha}{\beta} \frac{\omega}{r} \right)^\alpha L^{\alpha+\beta}$$

$$\begin{aligned} L^c(\omega, r, q) &= \left[\frac{q}{A \left(\frac{\alpha}{\beta} \frac{\omega}{r} \right)^\alpha} \right]^{\frac{1}{\alpha+\beta}} \\ K^c(\omega, r, q) &= \frac{\alpha}{\beta} \frac{\omega}{r} \left[\frac{q}{A \left(\frac{\alpha}{\beta} \frac{\omega}{r} \right)^\alpha} \right]^{\frac{1}{\alpha+\beta}} \\ &= \left(\frac{q}{A \left(\frac{\alpha}{\beta} \frac{\omega}{r} \right)^\alpha} \right)^{\frac{1}{\alpha+\beta}} \end{aligned}$$

$$C(\omega, r, q) = \omega L^c(\omega, r, q) + r K^c(\omega, r, q) = \left(\frac{q}{A} \right)^{\frac{1}{\alpha+\beta}} \left[\left(\frac{\alpha}{\beta} \frac{\omega}{r} \right)^{-\frac{\alpha}{\alpha+\beta}} + \left(\frac{\beta}{\alpha} \frac{r}{\omega} \right)^{-\frac{\beta}{\alpha+\beta}} \right]$$

$$L^c(\omega, r, q) = \left[\frac{q}{A \left(\frac{\alpha}{\beta} \frac{\omega}{r} \right)^\alpha} \right]^{\frac{1}{\alpha+\beta}} \quad K^c(\omega, r, q) = \frac{\alpha}{\beta} \frac{\omega}{r} \left[\frac{q}{A \left(\frac{\alpha}{\beta} \frac{\omega}{r} \right)^\alpha} \right]^{\frac{1}{\alpha+\beta}}$$

$$C(\omega, r, q) = \omega L^c(\omega, r, q) + r K^c(\omega, r, q)$$

$$A = \omega = r = 1, \quad \alpha = \beta = 1/2$$

$$L^c(q) = q \quad K^c(q) = q$$

$$C(q) = \omega L^c(q) + r K^c(q) = q + q = 2q$$

$$MC(q) = 2$$

$$AC(q) = \frac{C(q)}{q} = 2$$

$$L^c(\omega, r, q) = \left[\frac{q}{A \left(\frac{\alpha}{\beta} \frac{\omega}{r} \right)^\alpha} \right]^{\frac{1}{\alpha+\beta}} \quad K^c(\omega, r, q) = \frac{\alpha}{\beta} \frac{\omega}{r} \left[\frac{q}{A \left(\frac{\alpha}{\beta} \frac{\omega}{r} \right)^\alpha} \right]^{\frac{1}{\alpha+\beta}}$$

$$C(\omega, r, q) = \omega L^c(\omega, r, q) + r K^c(\omega, r, q)$$

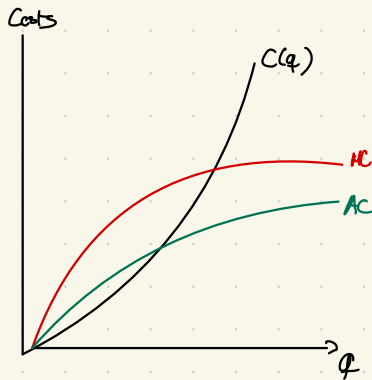
$$A = \omega = r = 1 \quad \alpha = \beta = 1/3$$

$$L^c(q) = q^{1/2} = q^{3/2} \quad K^c(q) = q^{3/2}$$

$$C(q) = 2q^{3/2}$$

$$\underline{MC(q)} = 2 \cdot \frac{3}{2} q^{1/2} = 3\sqrt{q}$$

$$\underline{AC(q)} = \frac{2q^{3/2}}{q} = 2\sqrt{q}$$



Short Run Cost minimization problem

$$\min_{K,L} wL + rK \quad \text{subject to: } q = F(K, L), \quad K = K_0$$

In this problem, input prices w and r , as well as q , and K_0 are treated as parameters or **exogenous variables**. The Technology F is also fixed. Notice that at least one input of production, in this case K , is fixed (at K_0 in this case).

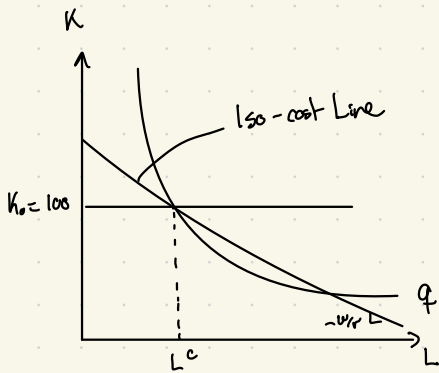
- We call the optimal solution(s) the **conditional short-run input demand(s)**: $L^c(w, r, q, K_0)$.
- In the two input case, The fact that we have the short-run restriction K_0 pins down costs.
- Notice that short-run cost, are in general, **not optimal with respect to the long run production plan**.

$$\text{Short Run Cost} = w \underbrace{L^c(w, r, q, K_0)}_{\text{Depends on the level of } q} + \underbrace{r K_0}_{\text{Doesn't depend on } q} = VC(q) + FC$$

Example: $q = A K^{1/2} L^{1/2}$

Say in the Short-Run $K_0 = 100$ units, $r = 1$

If the firm wants to produce q units how much labor it should use?



$$q = A (100)^{1/2} L^{1/2}$$

$$\Rightarrow q = A 10 L^{1/2}$$

$$\Rightarrow L_{SR}^c = \left(\frac{q}{10A} \right)^2$$

$$SRTC(q, w, r=1, K_0=100)$$

$$= w \underbrace{\frac{q^2}{100A^2}}_{VC(q)} + \underbrace{100}_{FC}$$

$$w=1, A=1$$

$$SRTC(q) = \frac{q^2}{100} + 100$$

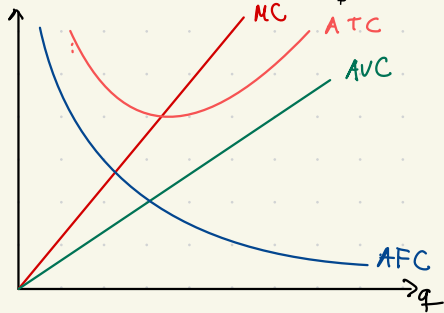
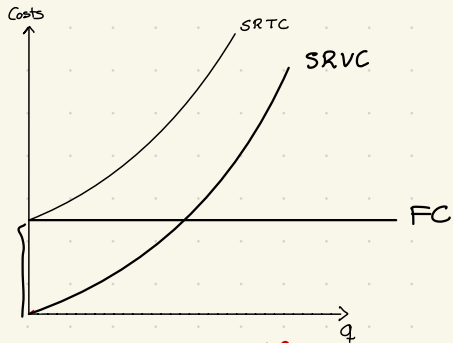
$$SRTC(q) = \frac{q^2}{100} + 100$$

$$MC(q) = \frac{q}{50}$$

$$ASRTC(q) = \frac{q}{100} + \frac{100}{q}$$

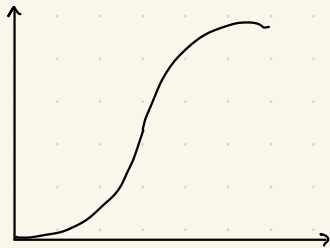
$$SRVC(q) = \frac{q^2}{100}, \quad SRFC(q) = 100$$

$$AVC(q) = \frac{q}{100}, \quad AFC(q) = \frac{100}{q}$$



Example: Cubic Costs

Recall: cubic production function



gives rise
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